

SYNTHESIZING MEANING



CREDITS

Narrated, curated and composed by Lisa-Marleen Mantel and Laura Wagner. In collaboration with DiscoDiffusion, PyTTI and StyleGAN synthesized dreams. Starring Mayssa Kaddoura as the voice of an advanced AI and Bessie Normand as the body of a human sensing device. Music «No more light» by Jimmy Svensson.

Master of Integrated Design at Köln International School of Design (KISD), part of TH Köln – University of Applied Sciences, Supervised by Prof. Nina Juric, Prof. Dr. Laura Poppiow and Prof. Dr. Lasse Scherffig. Published 31st January 2022.

SOFTWARE

PyTTI5-beta notebook (access for patrons of sportstrace48), Recovery DiscoDiffusion v4 Preview Alpha w/ Zooming and Chigozieni keyframes (both Colab notebooks are built upon the original notebook of Katherine Crowson), StyleGAN3, StyleGAN2-ada-Pytorch, RIFE, realESRGAN, Topaz Labs, Blender, Adobe Premiere, Adobe After Effects, Adobe Audition

HARDWARE

GPU NVIDIA P100 (through Google Colab) NVIDIA RTX 3070 (StyleGAN3 training and inference) NVIDIA RTX 3070 (StyleGAN3 training and inference)

An experimental short film produced with a custom designed production pipeline for generative Machine Learning to mediate speculative concepts of disembodied intelligence by Lisa-Marleen Mantel and Laura Wagner

In times of ubiquitous computing, artificial intelligence, and virtual humans, it seems like an obvious next step for technologists to see their chance for immortality in technological advancements. These future narratives are almost exclusively based on cartesian notions of disembodied intelligence. Influenced by posthumanist theorists, such as Rosi Braidotti, Donna Haraway, and Katherine Hayles, transhumanist approaches that envision the human body enhanced and augmented with machine parts are not only questioned but appropriated and inverted with the use of speculative design.



text_prompt = [A portrait of a woman with a nudibranch attached to her head | background is clean | by james jean | ernst haeckel | by neri oxman-]



text_prompt = [Hands are turning | each finger looks like a sensory device | LED lights are blinking at the end of each finger | painting by james jean:5 | illustration by ernst haeckel:1,]



MAKING - OF

Interview conducted by Prof. Nina Juric

NJ What makes your work outstanding, unique and special?

LM In this work posthuman approaches have been incorporated into a speculative design concept adapted from technological developments in cognitive science and artificial intelligence. It addresses the critique of transhumanist attempts of creating the superhuman and the continuation or amplification of the mind-body problem. Influenced by posthumanist thinkers it assumes a speculative future in which humans live in a

symbiotic, hybrid existence with other species such as AI. In contrast to focusing on perfecting and increasing the efficiency of the human species, the speculative concept developed during

the research focuses on a future AI that utilizes the human body to gain an embodied understanding of its environment. Transhuman concepts in which the human body is technologically enhanced are appropriated and inverted, and the human body itself becomes an enhancement in the role of a sensing device. Special about the creation process was the fact that we appropriated the very disruptive technologies that our theoretical concepts dealt with and used them as design tools for producing »Borrowed Limbs«. In our collaborative design workflow with AI tools we ourselves performed as the symbiotic, embodied link we describe in our speculative plot. Our organic bodies, capable of multimodal interaction, enhanced the image synthesization with context and subjective meaning. A constant

The short film »Borrowed Limbs« aims to emphasize the mediating role speculative design has in the face of disruptive technological advancements, as it allows for the exploration and simulation of ethical issues and provokes discussion about current developments and how they will influence our future way of life. The theoretical work results in a short film created with a designerly appropriation of AI tools. The suggested production pipeline, developed in the process, uses a myriad of AI tools. The main part of the footage was produced with a novel machine-learning-backed process called »CLIP guided diffusion« that allows for image synthesization based on language prompts that are carefully refined in a human-machine loop of communication. In a way, the practical processes involved in making the film reflect on the speculative ideas conveyed in it: the protagonist is an AI that utilizes a human body to gain an embodied understanding of its environment, a result of consequent machine-human hybridization – a posthuman being.



NJ What would be the next steps now for this work body and for your career as integrated design masters?

LM A part of the future role of designers will be to understand their profession as a discipline of thought experiments. When technological progress is accelerating, and the possibilities of appropriation are given, it is even more important to draw attention to the possible dangers of future developments. Since the tools we use as human beings are shaping our bodies and mindset of tomorrow we feel it is important to share our insights and the potential dangers with other creatives and digital creators. We are planning a series of workshops in which we would like to give an introduction to the design pipeline we have developed. Content will be the creative workflow where the machine is the creator and the human the curator. But also, and more importantly, a critical look at the foundation these technologies are built on.

NJ Let's do some foresight: Where do you see AI development in the context of audio-visual aesthetic production processes going right now?

LM Repetitive tasks have always been automated with technology, which is not new – but with AI, designers are gaining a tool that can be used to co-create in a drastically new way. A promising approach would be to view AI as a collaborator rather than a competitor, a colleague of a new species. Designers will be able to outsource tedious tasks and engage with machines collaboratively, generating and curating from an infinite latent space of possible designs. As a machine whisperer, the designer will enter a constant dialogue in setting the goals, parameters, and constraints before reviewing and curating the results. Concluding, adjusting, and repeating a generative process before the result can be manually refined will become an important task. AI enables many industries to move away from the prerogative of a one size fits all standardization approach for their products. A new approach to radical personalization is emergent. This could open up the possibility for designers to develop entirely new types of experiences. Concept visualization in character and product design, image-diffusion based



text-prompt brushes in software like Photoshop, AI texture packs, Realtime AI rendering in video games – these developments will change the way designers work.

For the creation of our short film we used the open-source version of DALL·E. After OpenAI released some of the code of DALL·E last year, many hobbyists set out to recreate it. Results of that are PyTTI and VQGAN + CLIP. With DALL·E 2 being released a few weeks ago it becomes apparent that rapid, market-driven implementation and industrial agendas are the main focus. The supposed improvements are based on the proximity to photorealism. These developments are also influential on the aesthetics and the presumed creative freedom these tools promise.

NJ Which specific topics and aspects of the Machine Learning realm would be crucial to integrate in the Integrated Design program or Code & Context program in near future? And why?

LM Exploring AI tools in a designerly way leads to an exploration of a component of AI frameworks from which behaviors emerge: Training data. Most datasets are web scraped – programmatically copied from the internet, and thus they inherit and carry forward biases from their birthplace. (The DALL·E 2 tests by Arthur Holland Michel visualize this fact very figuratively.) This dilemma is exacerbated by the fact that AI is introducing artificial content into the vast corpus of our collective representation of online knowledge, the internet. Some researchers and developers involved in AI technologies follow the dictum that AI can overcome all its limitations if fed with enough raw data; no or little effort is needed to make the AI system more sophisticated. Data omission and simplification for the sake of computational feasibility introduce and amplify bias even more. Once a model is trained with biased data, it serves as a multiplier. Machine Learning models trained today will inadvertently determine what future content will look like, as they continuously add new data to our collective internet knowledge, which in turn will be used to train new models.

Foreseeing the developments and integration of those tools in the industry it is crucial for designers to understand what data the tools they are using are built upon. This way misuse and discriminating results can be altered during the creation process.



text_prompt = [A portrait of a woman with a nudibranch attached to her head | background is clean- | by james jean | by ernst haeckel-]



text_prompt = [two people are hugging | their bodies are melting together in a bright light explosion | background is clean | Magic Realism Of Rob Gonsalves :1 | paintings by Jim Burns:2 | illustrations by ernst haeckel:1-]

LISA-MARLEEN MANTEL (she/her) completed her Master's degree (M.A. Integrated Design) at KISD. Previously, she studied Media Design (B.A.) at the DHBW Ravensburg. Her multidisciplinary experience derives from 7+ years of working in the field of creative technologies as an Art Director. In a current teaching assignment at KISD, she is working on the future curating role of designers in collaboration with Artificial Intelligence. www.lisamarleen.de

LAURA-MARLEEN WAGNER (she/her) completed her Master's degree (M.A. Integrated Design) at KISD. Previously she studied Communication Design (B.A.) at the FH Aachen. Since the beginning of this year she has been working as a research associate and interdisciplinary designer at KISD. Her research is directed at reconciling disembodied technologies with those aspects of the design profession that rely on tacit and embodied knowledge. www.lwgnr.de

AS AN ARTISTIC DUO they conduct workshops, teaching designers how to appropriate new technologies in the development stage, to anticipate disruptive changes to the design industry proactively. Their joint approach is using speculative design to reflect on possible future scenarios following disruptive technological advancements. Through their research they developed a specific interest in generative image synthesization with machine learning and discourses challenging cartesian notions of disembodied intelligence and AGI.



TRANS HUMANISM



There is an inherent urge in **human** beings to feel special. Although the cosmic perspective undoes this urge, the depictions of being the coronation of creation echo in humanist societies. Unfortunately, they seem not entirely flawless: the human body is transience. In religious contexts, this flaw is weakened by the promise of paradise after death. However, in times of ubiquitous computing, artificial intelligence, and virtual **humans**, it seems like an obvious next step for techno-optimists to see their chance for immortality in technological advancements.

FUTURE OF HUMANITY

The Cryonics Institute² created a business model for wealthy **humans** freezing their bodies. In tanks filled with liquid nitrogen, they are stored until medical technology can revive them in an unknown future. The Moravec Mind Transfer offers a complete detachment of the mortal body³. Hans Peter Moravec is an Austrian futurist who worked in the field of robotics and artificial intelligence. In his book *Mind Children* (1988), he suggests transforming the **human** mind into a mechanical structure of Nanobots. The nanobots are copies of the neurons of a human brain simulating it at the end of the process entirely, uploaded on a computer. Such transhuman approaches aim to enhance the **human** and secure the hegemony. Driven by dualistic beliefs⁴, those concepts mainly focus on enhancing parts of the **human** body mechanically or erasing the body entirely and only preserving the brain. Transhumanism as a notion is the fulfillment of the enlightenment project, which was mainly the perfectibility of the **human** through the application of scientific reason situated in the seveneenth century in Europe. One of the most famous representatives is philosopher and mathematician René Descartes, who hierarchized the mind and body, assuming that only humans can think. And not even among the human species, everybody is considered **human** by Cartesian male theorists. This designation is reserved for heterosexual, white cis males. Not included are those whom contemporary philosopher and feminist Rosi Braidotti calls the missing people⁵: among other women, Black, First Nation, dark- and darker, African, indigenous, LGBTQ+, neurodiverse, and people with disabilities. Moreover, these same people are consequently forgotten to transform into a transhumanist society. In the words of Braidotti, Transhumanism is nothing else than Neo-Humanism⁶. The Mind over Body hierarchy is also a widespread belief and argument while developing future digital tools. The mainly male-led digital field has a massive

In accordance with posthumanist thinkers such as Rosi Braidotti, Donna Haraway, and Katherine Hayles, we distinguish how we use the terms **human** and **human** (written in italics). In a humanist societal context, **human** is not to be understood as a neutral term.⁷ It has historically been used to refer exclusively to white, hetero, cis males and excludes so-called missing people (among others women, Black, First Nation, dark- and darker, African, indigenous, LGBTQ+, neurodiverse and people with disabilities). However, in a biological and evolutionary context, we refer to the species *homo sapiens sapiens* as human.

VS.

influence on technological advancements, making technology, not at all neutral. Cartesian theories are extended to the field of information technology. The transhumanist views create a basis that allows **human** beings to be seamlessly transferred into intelligent machines. There is no difference between: »bodily existence and computer simulation, cybernetic mechanism and biological organism, robot teleology, and **human** goals«⁷. In this modern dichotomy, white, European men are disembodied minds, whereas others are overexpressed or suppressed⁸. Dualism is used as an instrument of governance. Missing people who cannot participate in the **human** enhancement« will be erased just like the body of the enhanced ones. We are already augmenting, organizing, and externalizing our cognitive processes with machines. We store and query our knowledge and memories in ones and zeros and extend our perception with sensory machines that tell us in human-readable form meteorological forecast. We repair ourselves with artificial hearing aids and pacemakers. Human societies will be hyperconnected in the future, and the organic will become even more digitized. Conclusively **humans** could repeat oppression, exploitation, and inequality already seen in history by becoming non-biological thinking beings. Nevertheless, the exponential acceleration of the digital world is amplifying these significantly⁹.

QUESTIONING THE CARTESIAN WAY OF THINKING

A school of thought that sees itself as an abolishment of the body-mind dualism is posthumanism. It understands itself as a chance for those not seen as equally **human** and aims to overcome sexual, racial, and ethnic inequalities retrieved by anthropocentric history and perpetuated by transhumanist supporters. Instead of adapting, the individuals neglected by humanist structures should form transspecies alliances and companionships¹⁰. Yeon Jeong Gu, Soongsil University in Seoul, suggests reconfiguring a technologically mediated body as a new form of subjectivity¹¹. Influenced by posthuman thinkers like Rosi Braidotti and Katherine Hayles, her vision is a human body diversified and hybridized with objects, animals, or nature through technology. In this context, hybridity can also be understood as an intersection with other organisms. An important aspect is coexistence and symbiosis with other living beings. All this contrasts with an understanding of the world which puts one species in the center. Donna Haraway and her figures of thought like cyborgs and chimeras create a blueprint for this posthuman body. With her theoretical approaches of hybrids of machines and organisms¹², she declares that humans already fused with technology when using objects as tools. For philosopher Bernard Stiegler, objects are

indispensable for transporting human knowledge. They outsource thoughts from our bodies into non-human objects¹³. Through written language or art, experiences are passed on to the next generations. By using and integrating these tools into the human body, people reshape themselves and transform. Humanity and its capabilities multiply as media becomes increasingly complex. People have a reason because they use writing and art, not the other way around. The cultural scientist Zoë Sofoulis, on the other hand, criticizes the terms »posthuman« and »non-human objects« because they obscure the human labor invested in the creation of hybrid artifacts. The prefix »post«, in her opinion, is meant to evoke newness. She proposes the term »parahuman« instead to signify that action and intelligence exist just as much outside and alongside the human. Humanity is not redundant.¹⁴ This thesis continues to use the term »posthuman« but recognizes this criticism. The hybridity of organisms and objects is a preferable framework to consider transformation. The human body is in constant flux, interacting with other beings and objects. This process forms and reconstructs it constantly. Thousands of years of human history affected human behaviors and thoughts.¹⁵ From a socio-technical point of view, human personalities are often defined by connecting with technical or non-human entities. Deities or influential personalities are usually not shown alone or only naked in ancient or medieval depictions but in connection with objects such as a sword or book.¹⁶ On the contrary of technology-centered discourses where materiality must be transcended, technology is a part of an organism's body and shapes it. Posthuman identities require an exchange with each other but furthermore with non-human actors. Human thinking is related to the human body. This fact makes it crucial to practice interrelationships and value a symbiotic coexistence. Yeon Jeong Gu suggests experimenting with »technically mediated variant subjectivity beyond the categories of normality.«¹⁷ Katherine Hayles believes that future development should be conducted with care. Information like that of humanity cannot exist detached from a body. Embodiment gives context to information and meaning while it can be destroyed but not restored. The material world is fragile and cannot be replaced.¹⁸ Posthumanism signals not the end of humanity but the end of a specific concept of the **human** being. It was a concept that at best applied to that part of humanity that had the wealth and power to conceive of themselves as autonomous beings.¹⁹ Social and cultural structures need to be rethought with the **humanness** being decentered and focus on methods of communication and living through the other where boundaries are increasingly blurred.

Text: Lisa-Marleen Mantel, Laura Wagner / Reedited excerpt from their Master Thesis "Synthesizing Meaning – Designing a production pipeline for the artistic appropriation of generative Machine Learning to mediate speculative concepts", 2022

- ¹ Braidotti, Rosi, Hlavajova, Maria and contributors, »Posthuman Glossary.« Posthuman Critical Theory, London: Bloomsbury Academic, 20182 »Cryonics Institute« accessed January 22, 2022, <https://www.cryonics.org/>.
- ² »Cryonics Institute«, accessed January 22, 2022, www.cryonics.org.
- ³ N. Katherine Hayles, *How We Became Posthuman: Virtual Bodies in Cybernetics, Literature, and Informatics* (Chicago, Ill: University of Chicago Press, 1999), 1.
- ⁴ Ontological dualism states that body and mind are two forms of existence that are incompatible with each other and exist separately.
- ⁵ Rosi Braidotti and Maria Hlavajova and contributors, »Posthuman Glossary.« Posthuman Critical Theory, London: Bloomsbury Academic 2018, 342.
- ⁶ Rosi Braidotti, »A Theoretical Framework for the Critical Posthumanities.« Theory, Culture & Society 36, no. 6 (November 1, 2019): 32. www.doi.org/10.1177/0263276418771486
- ⁷ N. Katherine Hayles, »How We Became Posthuman«, 3.
- ⁸ Rosi Braidotti, The interview with Judith Butler (S. W. Oh, Trans. Korea). *Journal of Feminist Theories & Practices* (1999), 1, 278.
- ⁹ Yeon Jeong Gu, »The Disembodiment of Digital Subjects and the Disappearance of Women in the Representations of Cyborg, Artificial Intelligence, and Posthuman.« *Asian Women* 36, no. 4 (December 31, 2020): 33. www.doi.org/10.14443/1aw.2020.12.36.4.23.

¹⁰ Nicholas Gane, »When We Have Never Been Human, What Is to Be Done?: Interview with Donna Haraway.« *Theory, Culture & Society* 23, no. 7–8 (December 1, 2006): 135–58. www.doi.org/10.1177/0263276406069228

¹¹ Gu, »The Disembodiment of Digital Subjects and the Disappearance of Women in the Representations of Cyborg, Artificial Intelligence, and Posthuman.« 33.

¹² Nicholas Gane, »When We Have Never Been Human, What Is to Be Done?: Interview with Donna Haraway.« *Theory, Culture & Society* 23, no. 7–8 (December 1, 2006): 135–58. www.doi.org/10.1177/0263276406069228.

¹³ Patrick Crogan, »Bernard Stiegler: Philosophy, Technics, and Activism.« *Cultural Politics* 6, no. 2 (July 1, 2010): 133–56. www.doi.org/10.2752/175174310X12672016548162.

¹⁴ Zoë Souflis, »Post-, Nicht- und Parahuman«, in »Future Bodies. Zur Visualisierung von Körpern in Science und Fiction«, ed. Marie-Luise Angerer, Kathrin Peters and Zoë Sofoulis. 273–300, Wien, New York: Springer-Verlag 2002, 283-284.

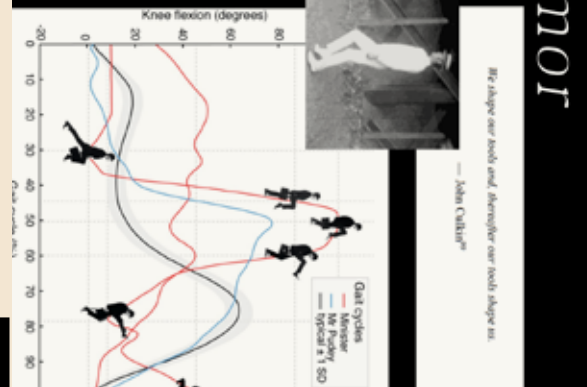
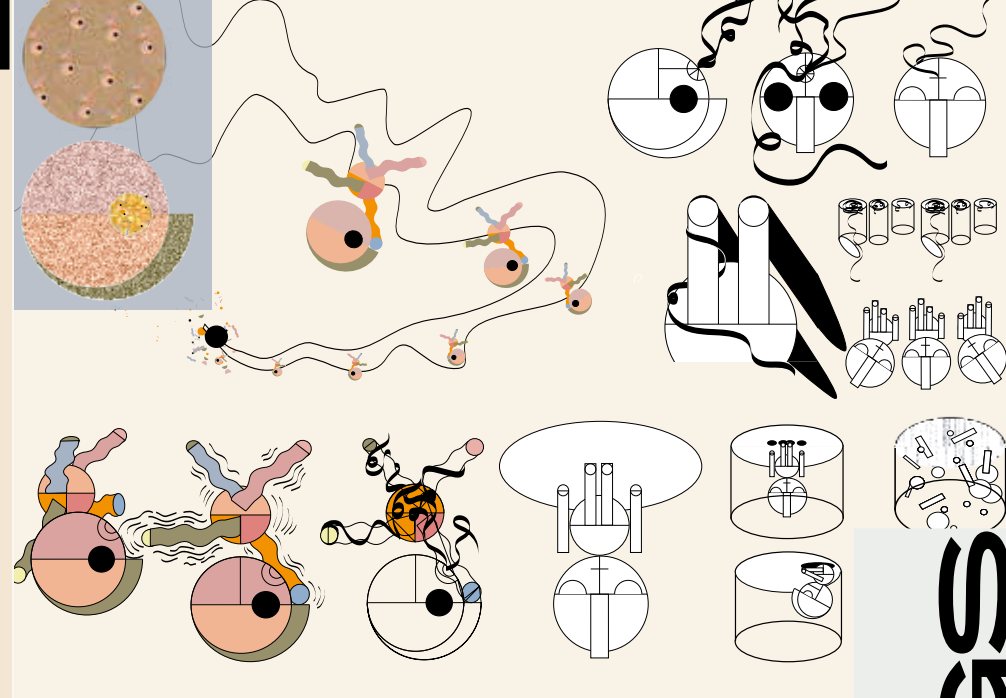
¹⁵ N. Katherine Hayles, »How We Became Posthuman«, 284.

¹⁶ Zoë Souflis, »Post-, Nicht- und Parahuman«, 295-296.

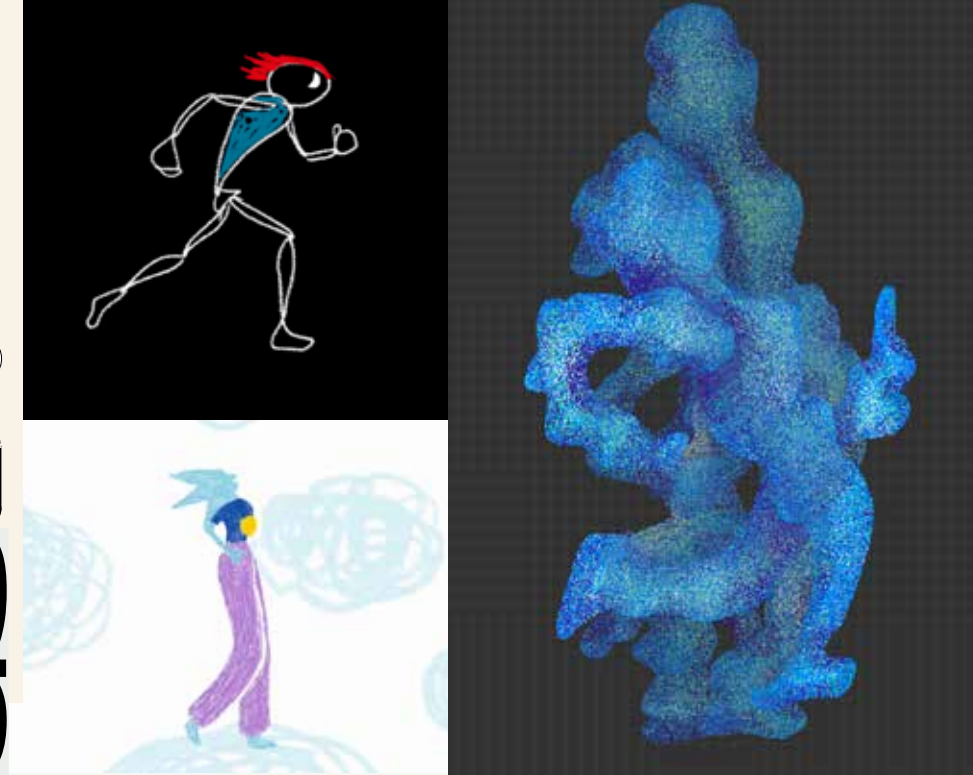
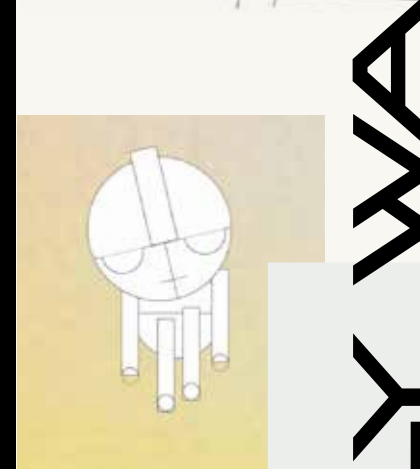
¹⁷ Gu, »The Disembodiment of Digital Subjects and the Disappearance of Women in the Representations of Cyborg, Artificial Intelligence, and Posthuman.« 42.

¹⁸ N. Katherine Hayles, »How We Became Posthuman«, 49.

¹⁹ N. Katherine Hayles, »How We Became Posthuman«, 49.



- SOB Karl Rogel
- NOUR Lars Köppl
- LIMBO Bessie Normand
- R-HA Valentina Stahrke
- ARDEN Jennifer Zachold
- FARRAH Natália Mota
- MURUMURU Gabriela Bornatto
- FLUIDO Bessie Normand
- BRILLY- B Amelia Sánchez



In this project students dealt with the moving human body in comparison to lifeless »inanimate« objects. Walk cycles are »poetry in motion« and an essential tool for animating creatures. This was the starting point for the conception of cyborg characteristics and their narratives. The class examined over 10 weeks the concept of »animation of silly walks« while understanding deeper physicality in exercising analogue and digital ways. Investigating on movements and their processes, as well as going into digital creation through performing new embodiments were crucial steps. Capturing and analyzing their own enactments were used as base for character animation development. Performing with the human body derived creating soul and spirit through the expression of the physicality of an action.

The aim was to look at humans affection to lifelike processes as a stylistic device. One of our protagonists of this matter was movement as an expressive form. On a content level the investigations specifically address cyborg motion concepts. How to shape such »personalities« and how to animate their (anti-)types and movements? To bring the figures to life, sounddesign was indispensable. This mid-term journey resulted in 9 animated, audio-visual motion design studies with silly walking characters. The students were free to choose their techniques, styles, stories and sound according to their practice based research.

Due to the pandemic, there were motion exercises performed by students at home and captured and uploaded, as we only met online. These turned out as great motion references and conversation pieces for the character developments and were used as motion templates. For research reasons a data base of motion studies on walkcycles was created along this investigations. Find it on the website.



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SS2020 / ONLINE